

Studying the stars

People have been fascinated by stars and planets since prehistoric times, but for thousands of years they could only study them with their unaided eyes. In the 17th century, telescopes opened up new ways of seeing the universe, and astronomers have been building bigger and better instruments ever since.

Reflector telescope, 1724



WHAT IS A TELESCOPE?

Telescopes use a large lens or mirror to collect light, and a smaller lens called the eyepiece to create images of distant objects that are brighter and larger than those formed by our normal eyesight. The principle was discovered by the Dutch lensmaker Hans Lippershey around 1608.

Photo of Orion nebula, 1883



Infrared camera view, 2010

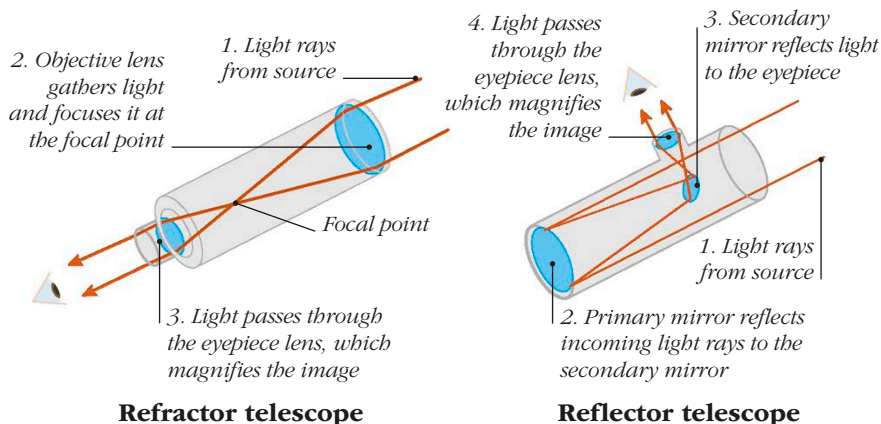


ASTROPHOTOGRAPHY

After the invention of photography in the 19th century, astronomers soon attached cameras to telescopes. Cameras can collect light over long periods, creating bright images that reveal much more detail than human eyesight can detect. Modern electronic cameras can detect types of light, such as infrared, that are invisible to human eyes.

TYPES OF TELESCOPE

Refractor telescopes were the first telescopes to be invented. They use large lenses to bend light rays so they cross at a focal point. Reflector telescopes use curved mirrors to bounce light rays to a focus. As the light rays spread apart again, one or more lenses in the eyepiece alter their paths to create a magnified, or enlarged, image.



► THE VERY LARGE TELESCOPE (VLT)

This is one of four huge telescopes built by the European Southern Observatory in Chile's high desert. Each telescope contains a mirror that is 26ft (8m) wide and weighs 24 tons. Using the VLT reveals images that are four billion times fainter than the unaided eye can detect.